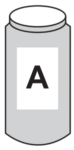
3. The following information is about hardness in water.

Hardness in water is caused by the presence of dissolved calcium (Ca^{2+}) and magnesium (Mg^{2+}) ions. These ions come from compounds that are dissolved by the water as it passes through rocks. Because they are dissolved, these ions are not removed during the water treatment processes.

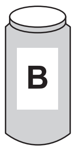
Hardness in water can be identified using soap solution. The relative hardness in different water samples can be determined by measuring the volume of soap solution required to produce a given lather.

Temporary hardness in water can be removed simply by boiling. Permanent hardness can be removed by passing the sample through an ion exchange column.

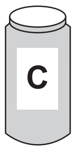
Three samples of water, **A**, **B** and **C**, were tested for hardness using soap solution. Distilled water was also tested.



Sample **A** required 8 cm^3 of soap solution to form lather before boiling.
It still required 8 cm^3 of soap solution to form lather after boiling.



Sample **B** required 12 cm^3 of soap solution to form lather before boiling.
It only required 6 cm^3 of soap solution to form lather after boiling.



Sample **C** required 15 cm^3 of soap solution to form lather before boiling.
It only required 2 cm^3 of soap solution to form lather after boiling.

Distilled water required 2 cm^3 of soap solution to form lather both before and after boiling.

Examiner only

(a) Which **one** of these statements best describes the water samples before boiling? Tick (✓) the correct answer. [1]

samples **A**, **B** and **C** contain magnesium or calcium ions ☐

only samples **A** and **B** contain magnesium or calcium ions ☐

only samples **B** and **C** contain magnesium or calcium ions ☐

none of the samples contain magnesium or calcium ions ☐

(b) What is the purpose of testing the distilled water? Tick (✓) the correct answer. [1]

to show the expected result for permanent hard water ☐

to show the expected result for temporary hard water ☐

to act as a control for the experiment ☐

to make sure the investigation is a fair test ☐

(c) In carrying out the tests on the samples, which variables need to be kept the same each time? Tick (✓) the correct answer. [1]

volume of soap solution and number of times shaken ☐

volume of water sample and volume of soap solution used ☐

volume of water sample and number of times shaken ☐

volume of water sample, volume of soap solution and number of times shaken ☐

(d) Which **one** of these statements best describes water sample **B**? Tick (✓) the correct answer and explain your choice. [3]

sample **B** contains temporary hardness only ☐

sample **B** contains permanent hardness only ☐

sample **B** contains a mixture of temporary and permanent hardness ☐

sample **B** does not contain hardness ☐

Explanation ....

....

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(e) In your opinion, do the advantages of living in a hard water area outweigh the disadvantages? Give **two** reasons to support your answer. [2]

....

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8

Examiner only

3. *(a)* The table gives the composition of six particles, **A**-**F**, which are either atoms or ions.

Particle	Number of protons	Number of neutrons	Number of electrons
A	14	14	14
B	19	20	18
C	15	16	18
D	16	16	16
E	11	12	11
F	12	12	10

(i) Which particles are atoms? Explain your choice. [2]

....

....

....

(ii) Which particles are positive ions? Give the charges on the particles you have chosen. [2]

....

....

(b) Carbon has two isotopes – carbon-12 and carbon-14.

Using these examples, explain what is meant by the term *isotope*. [2]

....

....

....

6

Examiner only

3. Some Year 10 students were given three unknown white solids – **A**, **B** and **C**.

They carried out a series of flame tests and silver nitrate tests to identify the solids.

Their results are shown in the table.

	Observations	
Solid	Flame test	Silver nitrate test
A	apple-green flame	cream precipitate
B	red flame	white precipitate
C	yellow flame	yellow precipitate

(a) Name solids **A**, **B** and **C**. [3]

A ....

B ....

C ....

(b) Complete and balance the symbol equation for the reaction between magnesium chloride and silver nitrate. [2]

$MgCl_2$ + $\square$ $AgNO_3$ $\longrightarrow$ ....

(c) 0.103 g of silver nitrate, $AgNO_3$, was used to make up a solution.

Calculate the number of moles of silver nitrate in this mass. Give your answer in **standard form**. [3]

$A_r(Ag) = 108$ $A_r(N) = 14$ $A_r(O) = 16$

Number of moles = ... mol

8

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3. The table shows the electronic structures of atoms of the elements **A–F**.

A–F are not the chemical symbols for the elements.

Element	Electronic structure
A	2
B	2,6
C	2,8,1
D	2,8,7
E	2,8
F	2,8,6

(a) Which of elements **A–F** is found in Group 6 and Period 3 of the Periodic Table?

Explain your choice, referring to electronic structure. [2]

....

....

(b) Which **two** of elements **A–F** are chemically inert?

Explain your choice, referring to electronic structure. [2]

....

....

(c) Element **D** has two isotopes. Isotope **1** has 18 neutrons and isotope **2** has 20 neutrons.

Complete the table by giving the atomic number and mass number of both isotopes. [2]

Isotope	Atomic number	Mass number
1		
2		

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(b) Tick (✓) **two** boxes which correctly describe the change in density and boiling point for the elements across Period 3. [2]

The density of metals and non-metals increases	☐
The boiling point of metals increases but the boiling point of non-metals shows no trend	☐
The density of metals shows no trend but the density of non-metals decreases	☐
The boiling point of metals and non-metals shows no trend	☐
The density of metals increases but the density of non-metals shows no trend	☐
The boiling point of metals shows no trend but the boiling point of non-metals decreases	☐
The density of metals decreases but the density of non-metals shows no trend	☐

(c) Argon is the next element in Period 3 after chlorine, Cl.

State why it is not possible to predict a melting point for argon using the information in the table. [1]

....

....

Examiner only

(d) Phosphorus is found in phosphoric acid, H_3PO_4.

(i) During the production of phosphoric acid, phosphorus is heated to 60 °C.

Give the state of phosphorus at 60 °C. Explain your choice. [2]

....

....

....

(ii) Phosphoric acid reacts with zinc to produce zinc phosphate and hydrogen.

Balance the equation for this reaction. [1]

$$\square\,Zn + \square\,H_3PO_4 \longrightarrow Zn_3(PO_4)_2 + \square\,H_2$$

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Examiner only

3. A food colouring contains a mixture of yellow and green dyes.

The table shows the R_f values of these dyes using different solvents in separate chromatography experiments.

Colour	R_f value when the solvent is		
	Water	Acid	Alcohol
yellow	0.74	0.35	0.00
green	0.76	0.81	0.46

(a) Why would yellow and green dyes be difficult to separate in a chromatography experiment using **water** as the solvent? [1]

....

....

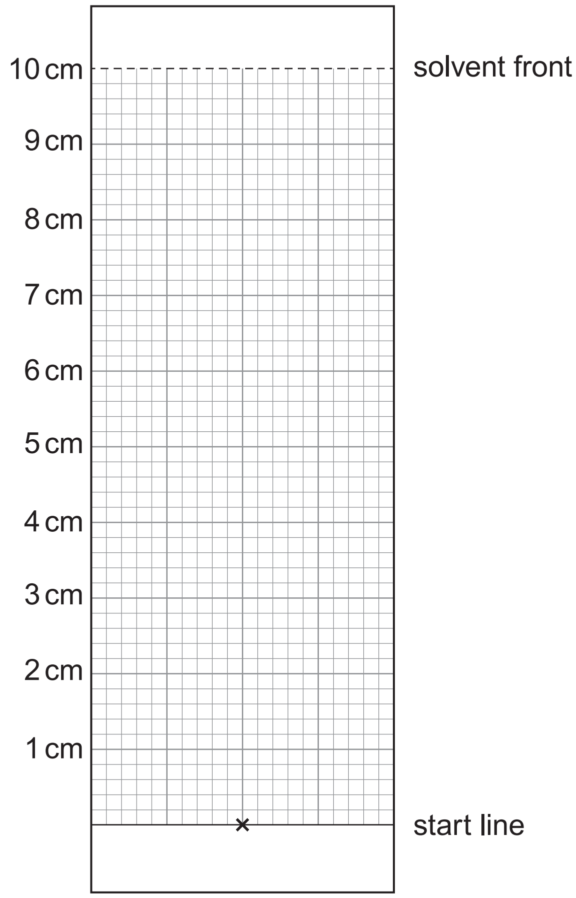
(b) Describe what you would expect to see in the chromatogram obtained using **alcohol** as the solvent. [1]

....

....

Examiner only

(c) Complete the chromatogram to show the results you would expect using **acid** as the solvent. [2]



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